# Sensor requirements and reliability

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Temperature could be omitted under a predefined benchmark rule when heart rate was available. A low overall error still concealed a long-session reliability gap.

| 0.978 | 6 vs 7 | 79.54% |
|---|---|---|
| primary test MAE* | required input fields | long-session interval coverage |

## A controlled comparison

The supplied 15,000-record CSV yielded 14,999 observations after one exact duplicate was removed. IDs were excluded from inputs. Fixed partitions contain 8,999 development, 3,000 calibration and 3,000 test records; five development folds are shared across comparisons.

A mean baseline, four Ridge pipelines and four XGBoost pipelines were compared. Each XGBoost input set received the same 30-configuration search. Calibration separates 1,500 interval-fit from 1,500 interval-audit records. Model selection and thresholds were frozen before a one-time final test release; test results did not retune the system.

## Frozen XGBoost results

| Available inputs | Fields | Test MAE* | Test RMSE* |
|---|---|---|---|
| Duration only | 1 | 11.508 | 16.953 |
| Basic measurements | 5 | 8.240 | 11.723 |
| Basic + heart rate (primary) | 6 | 0.978 | 1.576 |
| Full, including temperature | 7 | 1.032 | 1.645 |

*Against the same 3,000 recorded labels, in interpreted kcal/session. Original units and label construction are unverified.

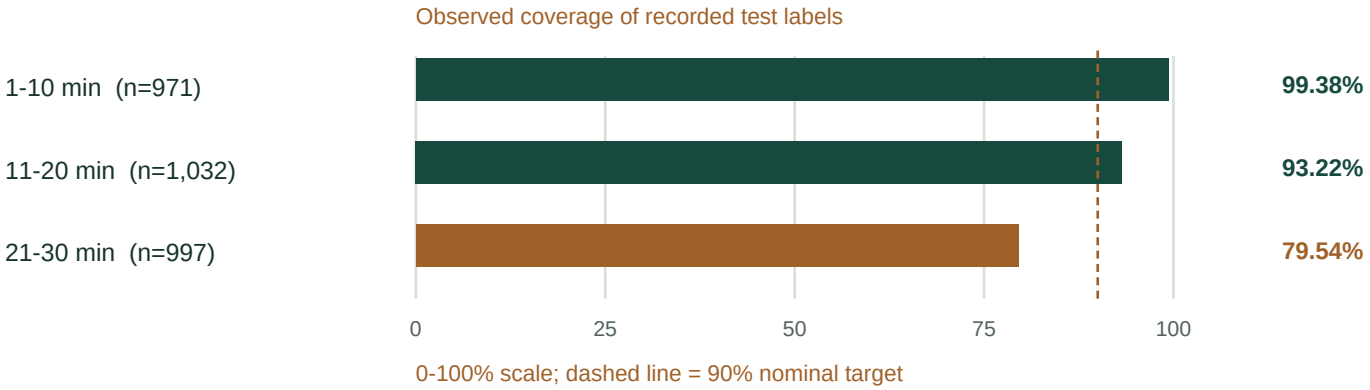
The heart-rate model's relative MAE change versus full inputs is **-5.22%**; approximate paired 95% percentile interval **[-6.52%, -3.82%]** using 2,000 row resamples. Its upper bound passes the project's +10% tolerance. Removing both sensors fails. One fewer field is an input-count result, not measured cost or time savings.

Scope: previously explored benchmark; unknown collection, possible participant dependence and unresolved dataset licensing. This is not verified physiological calorie measurement.

RELIABILITY / IMPLEMENTATION / EVIDENCE

# Make the failure cases visible

The primary fixed 90%-nominal interval covers 90.67% overall. Coverage differs substantially by session duration; an overall score does not establish reliability for every group.




For 21-30 minute sessions (997 records), coverage is only **79.54%**. Bounds remain unclipped; no subgroup recalibration followed the test. An earlier simulated Gaussian heart-rate-noise stress test (SD 5 bpm) raised development MAE from 1.080 to 7.777. That is a chosen simulation, not a measured device error distribution.

## A deployed, verifiable implementation

The free Hugging Face Static Space runs four exported frozen forests in the browser. Feature order, binary one-hot encoding, float32 comparisons and summation reproduce the Python pipelines. The app sends no prediction inputs to a prediction API and saves no submissions; hosting may retain asset-request logs.

Served files and repository hashes match the reviewed package. Downloaded JavaScript and models match Python predictions and unclipped bounds exactly on **5,854 synthetic sessions**, including split boundaries. The current source passes **82 Python tests and four JavaScript checks**. The author confirmed basic demo interaction; automated browser/mobile/accessibility checks were unavailable.

This demonstrates browser model delivery and deployment verification. A Python/Waitress Docker template is retained locally, but Linux container execution and hosted Docker deployment were not verified.

## Authorship and what is not measured

The author supplied the existing notebook/data, directed project choices, selected free hosting, completed account authentication and checked basic live interaction. Research code, evaluation tooling and documentation were produced with substantial AI-assistant support. Project ownership does not establish independent mastery of every implementation detail.

The usability study was skipped at the author's request: **no participant results or measured user benefit** exist. No clinical validity, adoption, health improvement, cost savings or new algorithm is claimed. Row bootstrap intervals assume exchangeability and omit training uncertainty. A documented measured-data replication would be a separate future study.

## Explore the evidence

Live demo: huggingface.co/spaces/Moobakr/calorie-lab-fewer-sensors
Existing repository: github.com/Moobakr/calories-burnt-prediction

Updated GitHub research source: publication pending. The accompanying source archive contains final evaluation, reliability and deployment verification reports, exact aggregate metrics and the authorship record.